Project 03 — OSPF Dynamic Routing

Multi-Area OSPF with BFD, MD5 Auth, IP SLA, and OSPFv3

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Platform: Cisco CML 2.9 | IOL Routers

Methodology: Build → Verify → Break → Fix

Setup and Scenario

Business Context

After completing Projects 01 and 02, the two-site HQ and Branch network relied entirely on manually configured static routes. Every subnet had to be explicitly listed on every router. There was no automatic failover — if a WAN link failed, traffic stopped. There was no authentication on routing updates, and convergence after a failure depended on the OSPF dead timer (40 seconds by default). This lab replaces the static routing foundation with a production-grade dynamic routing architecture.

What You Will Build

A three-router WAN with multi-area OSPF, adding a transit router (WAN-RTR1) as a second WAN path between HQ and Branch. The design includes:

- Single-area OSPF migration removing all static routes with zero downtime
- Multi-area design: Area 0 (WAN backbone), Area 1 (HQ campus), Area 2 (Branch)
- ABR route summarization: 10.1.0.0/16 at HQ-RTR1 and 10.2.0.0/16 at BR-RTR1
- MD5 authentication on all WAN-facing OSPF interfaces (key-id 1)
- OSPF cost manipulation: WAN-RTR1 path preferred (cost 20), direct link backup (cost 100)
- BFD at 300ms / multiplier 3 for sub-second link failure detection
- IP SLA ICMP probes with object tracking and floating static routes (AD 250)
- OSPFv3 replacing IPv6 static routes from Project 02
- Full link failure, convergence, and restoration testing

STAR Summary

Situation:

Two-site network with static routing only — no failover, no authentication, 40-second convergence on link failure, manual route maintenance for every subnet.

Task:

Migrate to multi-area OSPF with redundancy, authentication, sub-second BFD failover, and a tracked floating static last-resort backup layer.

Action:

Deployed WAN-RTR1 as transit router. Migrated to single-area OSPF then restructured into 3-area design with ABR summarization. Configured MD5 auth, cost-based path selection, BFD, IP SLA tracking, floating statics (AD 250), and OSPFv3. Diagnosed and resolved 5 real troubleshooting scenarios.

Result:

Network fails over in under 1 second (BFD). Three redundancy layers: preferred OSPF path via WAN-RTR1, backup OSPF via direct link, and tracked floating static last resort. Full IPv6 dynamic routing via OSPFv3.

Technologies Used

- OSPFv2 — multi-area (Areas 0, 1, 2) with loopback router IDs
- OSPFv3 — IPv6 dynamic routing replacing static routes from Project 02
- OSPF MD5 authentication on WAN interfaces (key-id 1, key: OSPF-WAN-KEY)
- OSPF cost manipulation — explicit ip ospf cost per interface
- OSPF passive-interface default with selective no passive on WAN links
- BFD — Bidirectional Forwarding Detection (300ms interval / multiplier 3)
- IP SLA — ICMP echo probes to directly connected /30 neighbor IPs
- Object Tracking — track ip sla reachability
- Floating static routes — AD 250 with track condition
- ABR route summarization — area range command
- ip ospf network point-to-point on all /30 WAN interfaces
- CDP verification before IP assignment on new devices

Network Topology

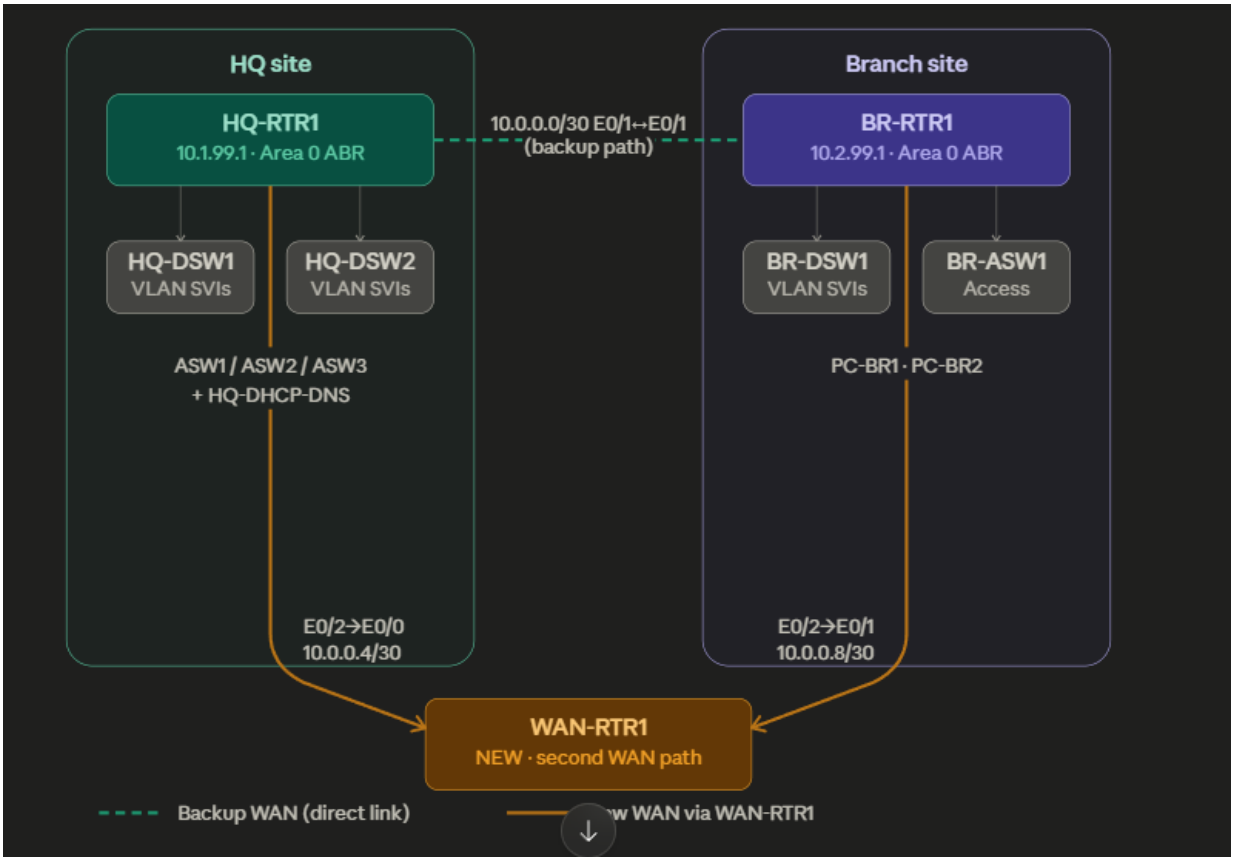

Figure T.1 — CML Topology — Project 03 OSPF Dynamic Routing (15 Nodes)

OSPF Area Design

Area 0 (Backbone): All three WAN /30 links and loopbacks. Area 1 (HQ Campus): HQ VLAN subinterfaces — summarized to 10.1.0.0/16 at ABR. Area 2 (Branch Campus): Branch VLAN subinterfaces — summarized to 10.2.0.0/16 at ABR. HQ-RTR1 and BR-RTR1 are ABRs. WAN-RTR1 is an internal Area 0 router only.

New Connections Added in Project 03

Side A	Interface	Side B	Interface	Subnet
HQ-RTR1	Ethernet0/2	WAN-RTR1	Ethernet0/1	10.0.0.4/30
WAN-RTR1	Ethernet0/0	BR-RTR1	Ethernet0/2	10.0.0.8/30

IP Addressing

WAN Links (Point-to-Point /30)

Link	Router A	IP	Router B	IP
Primary WAN	HQ-RTR1 E0/1	10.0.0.1/30	BR-RTR1 E0/1	10.0.0.2/30

Link	Router A	IP	Router B	IP
HQ to WAN-RTR1	HQ-RTR1 E0/2	10.0.0.5/30	WAN-RTR1 E0/1	10.0.0.6/30
BR to WAN-RTR1	BR-RTR1 E0/2	10.0.0.10/30	WAN-RTR1 E0/0	10.0.0.9/30

Loopbacks (Router IDs)

Router	Loopback0	OSPF Role
HQ-RTR1	10.0.255.1/32	ABR — Area 0 and Area 1
BR-RTR1	10.0.255.2/32	ABR — Area 0 and Area 2
WAN-RTR1	10.0.255.3/32	Internal — Area 0 only

HQ Campus VLANs (Area 1 — summarized to 10.1.0.0/16)

VLAN	Name	Subnet	Gateway
100	Engineering	10.1.100.0/24	10.1.100.1 (HQ-RTR1)
200	Sales	10.1.200.0/24	10.1.200.1
300	Guest	10.1.44.0/24	10.1.44.1
400	Server	10.1.40.0/24	10.1.40.1
999	Management	10.1.99.0/24	10.1.99.1

Branch Campus VLANs (Area 2 — summarized to 10.2.0.0/16)

VLAN	Name	Subnet	Gateway
100	Engineering	10.2.100.0/24	10.2.100.1 (BR-RTR1)
200	Sales	10.2.200.0/24	10.2.200.1
300	Guest	10.2.44.0/24	10.2.44.1
500	Wireless	10.2.50.0/24	10.2.50.1
999	Management	10.2.99.0/24	10.2.99.1

Pre-Work — Verify Physical Layer Before Any Config

Before touching OSPF, add WAN-RTR1 to the CML topology, cable it, and verify actual interface connections using CDP. Do NOT assign IPs until CDP confirms which physical interface connects to which neighbor.

Step 1: Add WAN-RTR1 to CML and run CDP

```
show cdp neighbors
show ip interface brief
```

> **WHY:** *In this build, the CML cables for WAN-RTR1 were connected to opposite interfaces from the design document. CDP revealed the actual connections before any IPs were assigned, preventing silent misconfiguration. Always trust CDP over the diagram.*

```
WAN-RTR1#show cdp neighbors
Capability Codes: R - Router, T - Trans Bridge, B - Source Route Bridge
                  S - Switch, H - Host, I - IGMP, r - Repeater, P - Phone,
                  D - Remote, C - CVTA, M - Two-port Mac Relay

Device ID        Local Intrfce     Holdtme    Capability  Platform  Port ID
HQ-RTR1.lab.local
                 Eth 0/1           157               R    Linux Uni Eth 0/2
BR-RTR1.lab.local
                 Eth 0/0           175               R    Linux Uni Eth 0/2

Total cdp entries displayed : 2
WAN-RTR1#
```

Figure P.0 — show cdp neighbors on WAN-RTR1 confirming actual cable layout

Step 2: Configure WAN-RTR1 base config (match IPs to actual cables)

```
hostname WAN-RTR1
!
interface Loopback0
 description ROUTER-ID-LOOPBACK
 ip address 10.0.255.3 255.255.255.255
!
interface Ethernet0/1
 description WAN-TO-HQ-RTR1-E0/2
 ip address 10.0.0.6 255.255.255.252
 no shutdown
!
interface Ethernet0/0
 description WAN-TO-BR-RTR1-E0/2
 ip address 10.0.0.9 255.255.255.252
 no shutdown
!
lldp run
```

Phase 1 — Single-Area OSPF Migration

Remove all static routes and replace with OSPF Area 0 on all three routers. Use passive-interface default to prevent OSPF hellos from reaching endpoints. Set all WAN interfaces to point-to-point to eliminate DR/BDR elections.

> **NOTE:** Critical sequence: configure OSPF on a router FIRST, wait for neighbors to reach FULL state, THEN remove static routes. This ensures zero downtime during cutover. Never remove statics before OSPF is established.

Step 1: OSPF on WAN-RTR1 (no statics to remove — start here)

```
router ospf 1
 router-id 10.0.255.3
 log-adjacency-changes detail
 passive-interface default
 no passive-interface Ethernet0/0
 no passive-interface Ethernet0/1
 network 10.0.255.3 0.0.0.0 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 10.0.0.8 0.0.0.3 area 0
```

Step 2: OSPF on HQ-RTR1 (configure first, remove statics after FULL)

```
router ospf 1
 router-id 10.0.255.1
 log-adjacency-changes detail
 passive-interface default
 no passive-interface Ethernet0/1
 no passive-interface Ethernet0/2
 network 10.0.255.1 0.0.0.0 area 0
 network 10.0.0.0 0.0.0.3 area 0
 network 10.0.0.4 0.0.0.3 area 0
 network 10.1.100.0 0.0.0.255 area 0
 network 10.1.200.0 0.0.0.255 area 0
 network 10.1.44.0 0.0.0.255 area 0
 network 10.1.40.0 0.0.0.255 area 0
 network 10.1.99.0 0.0.0.255 area 0
!
! After OSPF neighbors reach FULL:
no ip route 10.2.44.0 255.255.255.0 10.0.0.2
no ip route 10.2.99.0 255.255.255.0 10.0.0.2
no ip route 10.2.100.0 255.255.255.0 10.0.0.2
no ip route 10.2.200.0 255.255.255.0 10.0.0.2
```

Step 3: Fix DR/BDR election — apply to all WAN interfaces on all 3 routers

```
! HQ-RTR1 and BR-RTR1: Ethernet0/1 and Ethernet0/2
! WAN-RTR1: Ethernet0/0 and Ethernet0/1
interface Ethernet0/1
 ip ospf network point-to-point
interface Ethernet0/2
 ip ospf network point-to-point
```

WHY: *IOL Ethernet interfaces default to OSPF BROADCAST network type, triggering DR/BDR election on every /30 WAN link. With only 2 routers on a /30, the election is pure overhead and slows convergence. point-to-point eliminates it. Neighbors will briefly drop and reform — expected behavior during the change.*

Phase 1 Verification

```
show ip ospf neighbor        ! FULL/ - with Pri=0 on all routers
show ip route ospf           ! O routes present, no S routes remaining
show running-config | include ip route  ! Should return nothing
ping 10.2.100.1 source 10.1.100.1
```

```
HQ-RTR1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.0.255.3        0   FULL/  -        00:00:37    10.0.0.6        Ethernet0/2
10.0.255.2        0   FULL/  -        00:00:31    10.0.0.2        Ethernet0/1
HQ-RTR1#
```

Figure P.1 — show ip ospf neighbor on HQ-RTR1 — FULL/ - with Pri=0 on both neighbors

Phase 2 — Multi-Area OSPF Design

Split the single Area 0 into three areas. Move HQ campus VLANs to Area 1 and Branch campus VLANs to Area 2. Configure ABR summarization so WAN-RTR1 sees one summary per site instead of 5 individual /24 routes.

Step 1: HQ-RTR1 — move campus VLANs from Area 0 to Area 1

```
router ospf 1
 no network 10.1.100.0 0.0.0.255 area 0
 no network 10.1.200.0 0.0.0.255 area 0
 no network 10.1.44.0  0.0.0.255 area 0
 no network 10.1.40.0  0.0.0.255 area 0
 no network 10.1.99.0  0.0.0.255 area 0
 network 10.1.100.0 0.0.0.255 area 1
 network 10.1.200.0 0.0.0.255 area 1
 network 10.1.44.0  0.0.0.255 area 1
 network 10.1.40.0  0.0.0.255 area 1
 network 10.1.99.0  0.0.0.255 area 1
 area 1 range 10.1.0.0 255.255.0.0
```

Step 2: BR-RTR1 — move campus VLANs from Area 0 to Area 2

```
router ospf 1
 no network 10.2.100.0 0.0.0.255 area 0
 no network 10.2.200.0 0.0.0.255 area 0
 no network 10.2.44.0  0.0.0.255 area 0
 no network 10.2.50.0  0.0.0.255 area 0
 no network 10.2.99.0  0.0.0.255 area 0
 network 10.2.100.0 0.0.0.255 area 2
 network 10.2.200.0 0.0.0.255 area 2
 network 10.2.44.0  0.0.0.255 area 2
 network 10.2.50.0  0.0.0.255 area 2
 network 10.2.99.0  0.0.0.255 area 2
 area 2 range 10.2.0.0 255.255.0.0
```

WHY: *The area range command at each ABR generates a single Type 3 Summary LSA into Area 0. WAN-RTR1 sees O IA 10.1.0.0/16 and O IA 10.2.0.0/16 instead of 5 individual /24 routes per site. New subnets added within the range are automatically included with no changes required on WAN-RTR1 or the remote site.*

NOTE: HQ-RTR1 and BR-RTR1 will show 'O 10.x.0.0/16 is a summary, Null0' in the route table. This is correct behavior — IOS installs a discard route to Null0 at the ABR to prevent routing loops within the summary range. It is not an error.

```
WAN-RTR1#show ip route ospf
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

      10.0.0.0/8 is variably subnetted, 10 subnets, 3 masks
O        10.0.0.0/30 [110/110] via 10.0.0.10, 00:25:01, Ethernet0/0
                     [110/110] via 10.0.0.5, 01:16:48, Ethernet0/1
O        10.0.255.1/32 [110/11] via 10.0.0.5, 01:16:48, Ethernet0/1
O        10.0.255.2/32 [110/11] via 10.0.0.10, 00:25:01, Ethernet0/0
O IA     10.1.0.0/16 [110/20] via 10.0.0.5, 01:16:48, Ethernet0/1
O IA     10.2.0.0/16 [110/20] via 10.0.0.10, 00:25:01, Ethernet0/0
WAN-RTR1#
```

Figure P.2 — show ip route ospf on WAN-RTR1 — O IA 10.1.0.0/16 and O IA 10.2.0.0/16

Phase 3 — OSPF MD5 Authentication

Authenticate all OSPF routing updates on WAN interfaces. Without authentication, any device on a WAN link can inject false routes. MD5 means packets without the correct key are silently dropped. Both sides of every WAN link must use the same key ID (1) and password (OSPF-WAN-KEY).

Apply on all WAN interfaces on all 3 routers:

```
! HQ-RTR1 — Ethernet0/1 and Ethernet0/2
interface Ethernet0/1
 ip ospf message-digest-key 1 md5 OSPF-WAN-KEY
 ip ospf authentication message-digest
interface Ethernet0/2
 ip ospf message-digest-key 1 md5 OSPF-WAN-KEY
 ip ospf authentication message-digest
!
! Repeat identical config on BR-RTR1 E0/1 and E0/2
! and on WAN-RTR1 E0/0 and E0/1
```

WHY: *Passive interfaces do not send OSPF hellos so authentication on LAN interfaces is meaningless — there are no neighbors to authenticate. WAN links are the attack surface where a rogue device could be inserted. Interface-level MD5 is explicit and auditable per-interface. A mismatch in key ID or password drops the adjacency immediately.*

Verification

```
show running-config interface Ethernet0/1
! Look for: ip ospf authentication message-digest
!           ip ospf message-digest-key 1 md5 7 [encrypted-key]
!
show ip ospf neighbor   ! All neighbors still FULL/ -
```

Phase 4 — OSPF Cost Manipulation

Make path selection deterministic. Preferred path via WAN-RTR1 (total cost 20). Backup path via direct HQ-RTR1 to BR-RTR1 link (cost 100).

Path	Interfaces	Cost	Total	Role
Via WAN-RTR1	HQ-RTR1 E0/2 + WAN-RTR1 transit	10 + 10	20	PREFERRED
Direct HQ-BR	HQ-RTR1 E0/1	100	100	BACKUP

Apply on all 3 routers:

```
! HQ-RTR1
interface Ethernet0/1
 ip ospf cost 100   ! direct backup link
interface Ethernet0/2
 ip ospf cost 10    ! preferred WAN-RTR1 path
!
! BR-RTR1 (mirror of HQ-RTR1)
interface Ethernet0/1
 ip ospf cost 100
interface Ethernet0/2
 ip ospf cost 10
!
! WAN-RTR1 (transit router — both links preferred)
interface Ethernet0/0
 ip ospf cost 10
interface Ethernet0/1
 ip ospf cost 10
```

> **WHY:** *Explicit cost values are deterministic and portable. When you read the config you immediately know the cost — no math required. With bandwidth-calculated costs you must know the reference bandwidth and calculate. Explicit costs survive interface type changes without unexpected side effects.*

Phase 4 Verification

```
show ip route 10.2.100.1   ! Next hop: 10.0.0.6 (WAN-RTR1), metric 30
```

```
traceroute 10.2.100.1 source 10.1.100.1
! Expected: Hop 1: 10.0.0.6 (WAN-RTR1), Hop 2: 10.0.0.10 (BR-RTR1)
```

```
HQ-RTR1#traceroute 10.2.100.1 source 10.1.100.1
Type escape sequence to abort.
Tracing the route to 10.2.100.1
VRF info: (vrf in name/id, vrf out name/id)
  1 10.0.0.6 2 msec 1 msec 2 msec
  2 10.0.0.10 2 msec *  3 msec
HQ-RTR1#
```

Figure P.4 — traceroute HQ to Branch confirming path goes through WAN-RTR1

Phase 5 — Link Failure and Convergence Testing

Prove the backup path activates automatically. Shut the preferred WAN-RTR1 link, confirm OSPF reconverges to the direct backup, restore the link, and confirm traffic returns to the preferred path.

Step 1: Baseline — confirm preferred path active on HQ-RTR1

```
show ip ospf neighbor           ! Both neighbors: FULL/ -
show ip route 10.2.100.1        ! Via 10.0.0.6 on E0/2, metric 30
traceroute 10.2.100.1 source 10.1.100.1
```

Step 2: Simulate failure — shut HQ-RTR1 Ethernet0/2

```
configure terminal
 interface Ethernet0/2
  shutdown
end
```

Step 3: Verify failover — backup path active

```
show ip ospf neighbor
! WAN-RTR1 (10.0.255.3) should be gone
show ip route 10.2.100.1
! Expected: via 10.0.0.2 on Ethernet0/1, metric 110
```

```
HQ-RTR1#show ip route 10.2.100.1
Routing entry for 10.2.0.0/16
  Known via "ospf 1", distance 110, metric 30, type inter area
  Last update from 10.0.0.6 on Ethernet0/2, 00:27:01 ago
  Routing Descriptor Blocks:
  * 10.0.0.6, from 10.0.255.2, 00:27:01 ago, via Ethernet0/2
      Route metric is 30, traffic share count is 1
HQ-RTR1#
```

Figure P.5a — Route switches to backup path (metric 110) after E0/2 shutdown

Step 4: Restore E0/2 and verify preferred path returns

```
configure terminal
 interface Ethernet0/2
  no shutdown
end
!
show ip route 10.2.100.1
! Expected: back to 10.0.0.6 on Ethernet0/2, metric 30
```

```
HQ-RTR1#show ip route 10.2.100.1
Routing entry for 10.2.0.0/16
  Known via "ospf 1", distance 110, metric 30, type inter area
  Last update from 10.0.0.6 on Ethernet0/2, 00:31:31 ago
  Routing Descriptor Blocks:
  * 10.0.0.6, from 10.0.255.2, 00:31:31 ago, via Ethernet0/2
      Route metric is 30, traffic share count is 1
HQ-RTR1#
```

Figure P.5b — Route restored to preferred WAN-RTR1 path after E0/2 no shutdown

State	Next Hop	Metric	Interface
Normal	10.0.0.6	30	Ethernet0/2 (via WAN-RTR1)
E0/2 shutdown	10.0.0.2	110	Ethernet0/1 (direct backup)
Restored	10.0.0.6	30	Ethernet0/2 (preferred returns)

Phase 6 — BFD for Sub-Second Failover

OSPF's default dead timer is 40 seconds — a link failure can take up to 40 seconds to trigger reconvergence. BFD runs independently of OSPF and detects link failures in milliseconds. When BFD detects failure it immediately notifies OSPF.

Parameter	Value	Result
BFD interval	300ms	Probe sent every 300ms
min_rx	300ms	Accept probes every 300ms
Multiplier	3	Failure declared after 3 missed probes
Detection time	900ms	Sub-second — 44x faster than OSPF dead timer

Apply on all WAN interfaces on all 3 routers:

```
! HQ-RTR1 E0/1 and E0/2 (repeat for BR-RTR1 and WAN-RTR1)
interface Ethernet0/1
 bfd interval 300 min_rx 300 multiplier 3
 ip ospf bfd
interface Ethernet0/2
 bfd interval 300 min_rx 300 multiplier 3
 ip ospf bfd
```

Phase 6 Verification

```
show bfd neighbors
```

```
! Expected:
! NeighAddr   LD/RD   RH/RS  State  Int
! 10.0.0.2    1/1     Up     Up     Et0/1
! 10.0.0.6    2/2     Up     Up     Et0/2
!
show ip ospf neighbor   ! All still FULL/ -
```

```
HQ-RTR1#show bfd neighbors

IPv4 Sessions
NeighAddr                              LD/RD         RH/RS     State     Int
10.0.0.2                                3/1          Up        Up        Et0/1
10.0.0.6                                2/2          Up        Up        Et0/2
HQ-RTR1#
```

Figure P.6 — show bfd neighbors on HQ-RTR1 — all sessions Up/Up

Phase 7 — IP SLA + Tracked Floating Static Route

Add a third redundancy layer. If OSPF itself fails (not just a link), a tracked floating static route (AD 250) activates automatically. The IP SLA probe target must be reachable independently of OSPF.

> **NOTE:** Critical design lesson learned during this build: the initial probe targeted WAN-RTR1's loopback (10.0.255.3), reachable only via OSPF routes. When OSPF was shut down to test the backup, the probe lost its route, the track went Down, and the floating static never installed — defeating the entire backup mechanism. Fix: always probe the directly connected /30 neighbor IP. No routing needed — just ARP.

Step 1: Configure IP SLA, tracking, and floating static on HQ-RTR1

```
ip sla 10
 icmp-echo 10.0.0.2 source-ip 10.0.0.1
 timeout 1000
 threshold 1000      ! threshold must be <= timeout
 frequency 5
exit
!
ip sla schedule 10 life forever start-time now
!
track 10 ip sla 10 reachability
!
ip route 10.2.0.0 255.255.0.0 10.0.0.2 250 track 10
```

Step 2: Mirror on BR-RTR1

```
ip sla 10
 icmp-echo 10.0.0.1 source-ip 10.0.0.2
 timeout 1000
 threshold 1000
 frequency 5
exit
ip sla schedule 10 life forever start-time now
track 10 ip sla 10 reachability
ip route 10.1.0.0 255.255.0.0 10.0.0.1 250 track 10
```

> **WHY:** *A directly connected /30 IP requires only ARP — no routing protocol. The probe survives an OSPF failure because the physical E0/1 link is still up. AD 250 sits below OSPF (AD 110), so the floating static is invisible during normal operation and only installs when OSPF routes disappear from the table.*

Phase 7 Verification

```
show ip sla statistics     ! Successes incrementing, return code: OK
show track                 ! Reachability is Up
```

```
show ip route static       ! Empty — OSPF preferred (AD 110 < 250)
!
! Test with OSPF shut down on BR-RTR1:
! router ospf 1 -> shutdown
! show track            -> Reachability is Up (probe still works)
! show ip route static  -> S 10.1.0.0/16 [250/0] via 10.0.0.1
! ping 10.1.100.1 source 10.2.100.1 -> !!!!! 100%
```

```
BR-RTR1#show track
Track 10
  IP SLA 10 reachability
  Reachability is Up
    2 changes, last change 00:35:31
  Latest operation return code: OK
  Latest RTT (millisecs) 1
  Tracked by:
    Static IP Routing 0
BR-RTR1#show ip sla statistics
IPSLAs Latest Operation Statistics

IPSLA operation id: 10
        Latest RTT: 2 milliseconds
Latest operation start time: 02:49:40 UTC Sun Apr 12 2026
Latest operation return code: OK
Number of successes: 430
Number of failures: 0
Operation time to live: Forever


BR-RTR1#
```

Figure P.7a — show track and show ip sla statistics — Reachability Up, return code OK

```
BR-RTR1#show ip route static
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, m - OMP
       n - NAT, Ni - NAT inside, No - NAT outside, Nd - NAT DIA
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       H - NHRP, G - NHRP registered, g - NHRP registration summary
       o - ODR, P - periodic downloaded static route, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
       & - replicated local route overrides by connected

Gateway of last resort is not set

BR-RTR1#
```

Figure P.7b — show ip route static on BR-RTR1 with OSPF shut — S 10.1.0.0/16 [250/0] installed

Phase 8 — OSPFv3 for IPv6

Replace IPv6 static routes from Project 02 with OSPFv3 multi-area dynamic routing. IPv6 exists only on the direct HQ-BR WAN link (E0/1) and VLAN 100 subinterfaces. WAN-RTR1 backup path has no IPv6 in this design.

Step 1: Configure OSPFv3 on HQ-RTR1

```
no ipv6 route 2001:DB8:2:100::/64 2001:DB8:0:1::2
!
ipv6 unicast-routing
!
ipv6 router ospf 10
 router-id 10.0.255.1
!
interface Ethernet0/1
 ipv6 ospf 10 area 0
 ipv6 ospf network point-to-point
!
interface Ethernet0/0.100
 ipv6 ospf 10 area 1
```

Step 2: Configure OSPFv3 on BR-RTR1

```
no ipv6 route 2001:DB8:1:100::/64 2001:DB8:0:1::1
!
ipv6 unicast-routing
!
ipv6 router ospf 10
 router-id 10.0.255.2
!
interface Ethernet0/1
 ipv6 ospf 10 area 0
 ipv6 ospf network point-to-point
!
interface Ethernet0/0.100
 ipv6 ospf 10 area 2
```

> **WHY:** *OSPFv3 needs the same point-to-point treatment as OSPFv2. Apply ipv6 ospf network point-to-point on both sides before checking adjacency. A one-sided change produces a transient NET_TYPE_MISMATCH log message that resolves automatically once both routers are updated.*

Phase 8 Verification

```
show ospfv3 neighbor
```

```
! Expected: Neighbor ID 10.0.255.2, Pri 0, State FULL/ -
!
show ipv6 route ospf
! Expected: OI 2001:DB8:2:100::/64 [110/110] via FE80::..., Ethernet0/1
!
ping 2001:DB8:2:100::1 source 2001:DB8:1:100::1
ping 2001:DB8:1:100::1 source 2001:DB8:2:100::1
```

```
HQ-RTR1#show ospfv3 neighbor

          OSPFv3 10 address-family ipv6 (router-id 10.0.255.1)

Neighbor ID     Pri   State           Dead Time   Interface ID    Interface
10.0.255.2        0   FULL/  -        00:00:35    2               Ethernet0/1
HQ-RTR1#
```

Figure P.8a — show ospfv3 neighbor on HQ-RTR1 — FULL/ - with Pri=0

```
HQ-RTR1#ping 2001:DB8:2:100::1 source 2001:DB8:1:100::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:2:100::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:1:100::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms
HQ-RTR1#
```

Figure P.8b — IPv6 ping from HQ VLAN 100 to Branch VLAN 100: !!!!! 100%

Final Verification Checklist

#	Test	Command	Device	Expected
1	OSPF neighbors FULL	show ip ospf neighbor	All routers	FULL/ - Pri=0
2	No statics remain	show run | include ip route	HQ/BR-RTR1	Empty
3	O IA summaries on WAN-RTR1	show ip route ospf	WAN-RTR1	O IA 10.1.0.0/16 and 10.2.0.0/16
4	E0/2 cost = 10	show ip ospf int E0/2	HQ/BR-RTR1	Cost: 10, POINT_TO_POINT
5	E0/1 cost = 100	show ip ospf int E0/1	HQ/BR-RTR1	Cost: 100
6	MD5 auth on WAN ints	show run int E0/1 | include ospf	All routers	authentication message-digest
7	BFD sessions Up	show bfd neighbors	All routers	Up/Up on all sessions
8	IP SLA running	show ip sla statistics	HQ/BR-RTR1	Successes incrementing, OK
9	Track object Up	show track	HQ/BR-RTR1	Reachability Up
10	Traceroute via WAN-RTR1	traceroute 10.2.100.1 src 10.1.100.1	HQ-RTR1	Hop 1: 10.0.0.6
11	OSPFv3 neighbor FULL	show ospfv3 neighbor	HQ/BR-RTR1	FULL/ - Pri=0
12	IPv4 end-to-end	ping 10.2.100.1 src 10.1.100.1	HQ-RTR1	!!!!!
13	IPv6 end-to-end	ping 2001:DB8:2:100::1	HQ-RTR1	!!!!!

Troubleshooting Log — Project 03

Entry 1 — WAN-RTR1 Interface IP Mismatch

Symptom: Pings from WAN-RTR1 to HQ and Branch failing 0% immediately after config. Same-subnet /30 pings require only ARP — they should never fail.

Root Cause: CML cables were connected to opposite interfaces from the design. IPs assigned from the design had the wrong side on each interface. Additionally, 'no ip address' without the full mask left E0/0 unassigned.

Fix: Ran 'show cdp neighbors' to discover actual cable layout. Reassigned IPs to match physical connections confirmed by CDP. Pings returned 100% after fix.

Lesson: Always run 'show cdp neighbors' BEFORE assigning IPs to any new device. Never assume CML cable connections match the design diagram.

Entry 2 — DR/BDR Election on /30 WAN Links

Symptom: 'show ip ospf neighbor' showed FULL/DR and FULL/BDR instead of FULL/ - with Pri=0 on point-to-point /30 WAN links.

Root Cause: IOL Ethernet interfaces default to OSPF network type BROADCAST regardless of subnet mask. This triggers DR/BDR election on every link, including /30s with only 2 endpoints.

Fix: Applied 'ip ospf network point-to-point' to all WAN interfaces on all 3 routers. Adjacencies briefly dropped and reformed — expected during network type change.

Lesson: Always apply 'ip ospf network point-to-point' on /30 WAN links in IOL. The same applies to OSPFv3 — use 'ipv6 ospf network point-to-point'.

Entry 3 — Intentional Failover Test (Phase 5)

Symptom: Intentional outage: shut HQ-RTR1 E0/2 to verify OSPF reconverges to backup path without waiting 40 seconds.

Root Cause: Intentional test. No actual fault.

Fix: Interface-down triggered immediate adjacency loss. Route metric changed from 30 (preferred via WAN-RTR1) to 110 (backup direct link). On restoration, traffic returned to preferred path automatically.

Lesson: Interface-down events trigger immediate OSPF adjacency removal — no dead timer wait. Cost engineering produces deterministic failover.

Entry 4 — IP SLA Probe Would Not Start

Symptom: IP SLA not running. Track showed Reachability Down, return code Unknown. 'show ip sla statistics' returned Unknown counters. Operation time to live: 0.

Root Cause: IOS rejects 'ip sla schedule' when threshold > timeout. Default threshold is 5000ms; timeout was 1000ms. The schedule command was silently rejected, leaving the probe in notInService state.

Fix: Removed and recreated SLA with threshold 1000 and timeout 1000. After fix: track showed Reachability Up with RTT 1ms, success counters incrementing.

Lesson: When IP SLA will not start, check 'show ip sla configuration' for 'Pending trigger' and 'notInService'. Always set threshold equal to or less than timeout.

Entry 5 — IP SLA Probe Failed During OSPF Outage

Symptom: When OSPF was shut down to test the floating static backup, the track went Down. Floating static never installed. Network lost all inter-site reachability.

Root Cause: The probe target was WAN-RTR1 loopback 10.0.255.3 — reachable only via OSPF routes. When OSPF died, the route to the probe target disappeared, the probe failed, track went Down, and the floating static never installed.

Fix: Changed probe target to the directly connected /30 IP on the backup link (10.0.0.2 from HQ-RTR1, 10.0.0.1 from BR-RTR1). A /30 directly connected IP requires only ARP — it works regardless of OSPF state.

Lesson: IP SLA probe target must be reachable independently of the routing protocol it backs up. Always probe a directly connected interface IP on the backup path, never a loopback or remote address.

Break / Fix Challenge

Each challenge introduces a deliberate misconfiguration. Diagnose using only show commands — do not look at the running-config until you have a hypothesis.

Challenge 1 — OSPF adjacency drops after applying MD5

Symptoms: After adding MD5 authentication, one WAN adjacency drops and never recovers. 'show ip ospf neighbor' shows EXSTART or DOWN oscillation on one interface.

Hint: Check 'show ip ospf interface E0/x' on both sides. Compare the authentication type. Run 'debug ip ospf adj' briefly to see the mismatch message.

Challenge 2 — Traffic taking the backup path instead of WAN-RTR1

Symptoms: Traceroute from HQ to Branch shows only 1 hop (direct to BR-RTR1). The preferred WAN-RTR1 path should produce 2 hops. Both OSPF neighbors are FULL.

Hint: Check 'show ip ospf interface E0/1' and 'show ip ospf interface E0/2' on HQ-RTR1. Compare the Cost values. What was not applied?

Challenge 3 — IP SLA showing Unknown return code after scheduling

Symptoms: After configuring and scheduling IP SLA, 'show ip sla statistics' shows Return code: Unknown and Number of successes: 0. Track object shows Down.

Hint: Run 'show ip sla configuration 10'. Look at the Threshold and Timeout values. Which one is larger, and why does that matter?

Challenge 4 — OSPFv3 neighbor stuck at EXSTART

Symptoms: After configuring OSPFv3, 'show ospfv3 neighbor' shows EXSTART and never reaches FULL. IPv4 OSPF is working correctly on the same interfaces.

Hint: Run 'show ospfv3 interface E0/1' on both routers. Compare the network type. Is it the same on both sides?

Build date: 2026-04-12 **Platform:** Cisco IOL in CML 2.9 **Nodes:** 15 (adds WAN-RTR1 to Project 02 topology)